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Ayoub Khial

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PROFILE

Driven Computer Science student at Concordia University with a focus on AI Systems, DevOps, and Scalable Software Architecture. Experienced in architecting RAG-based legal platforms, fine-tuning LLMs, and automating secure deployment pipelines. Proven track record of leading technical teams from hackathon-winning concepts to functional MVPs. Skilled in bridging the gap between hardware (IoT), deep learning, and modern web infrastructure.

PROFESSIONAL EXPERIENCE

Phon.ia – AI-Driven Phonetics & Literacy Platform — **Montreal, Canada**
AI Developer & Project Coordinator — *October 2024–May 2025*

- Built an end-to-end AI driven literacy platform for adult French learners, taking ownership of system design, development, and delivery of a functional MVP within a 6 month research timeline.
- Coordinated a 4-person technical team by translating pedagogical requirements into concrete engineering tasks across AI, backend, and frontend components.
- Implemented speech recognition (STT) and text to speech pipelines (TTS) to evaluate phoneme pronunciation, reading fluency, and oral comprehension, providing real time corrective feedback to 100+ beta users.
- Designed logic combining rule-based evaluation and LLM assisted reasoning to assess reading comprehension and written responses, enabling adaptive difficulty progression per user.
- Architected a PostgreSQL backed backend handling secure authentication, user profiles, progress tracking, and persistent conversation history with strong data integrity guarantees.
- Designed and implemented an accessibility focused UI optimized for adult learners with learning difficulties, integrating local LLMs and external APIs while maintaining low latency interactions and improving user engagement by 30%.

LawMapAI – LegalTech Startup Initiative — **Montreal, Canada**
Co-founder & Full-Stack Developer — *May 2025–Current*

- Scaled a hackathon-winning prototype into a production-ready legal research platform, enabling lawyers to query, summarize, and analyze long documents (up to 400 pages each) with sub-second semantic search.
- Designed and implemented a custom RAG pipeline on individual PDFs, integrating embeddings, document chunking, and retrieval logic to provide accurate answers and contextual summaries.
- Built automated ingestion and parsing pipelines capable of processing hundreds of pages per document to extract case law, statutes, and citation relationships.
- Added a mind map visualization feature to simplify complex legal hierarchies and illustrate connections between cases, statutes, and user queries.
- Integrated optional web research functionality, allowing the system to augment PDF-based knowledge with external legal sources for richer answers.
- Developed the platform using Python Flask and React, maintaining clear separation between AI inference, data services, and frontend visualization to support maintainability, scalability, and rapid iteration.

CONTRACTS/PROJECTS

Neural Engine Architect (Transformer from Scratch) — **Aug 2024 – Nov 2024**

- Deep Learning Development: Engineered a custom Transformer architecture using PyTorch, implementing multi-head attention, positional encoding, and layer normalization from the ground up.
- Model Optimization: Developed custom training loops and hyperparameter tuning scripts to optimize model convergence. Focused on understanding the mathematical foundations of gradient descent and weight initialization to bridge the gap between AI theory and practical implementation.

BMO AI Emotional Assistant (Multimodal IoT Robot)
AI Dev – IoT integrator

Nov 2024 – Feb 2025

- Co-managed the development of a multifunctional chatbot robot with full IoT integration as part of a 4-person team, targeting an interactive and intelligent user experience.
- Designed a comprehensive 3D model and integrated advanced AI features, including LLM-based conversational capabilities and AI vision.
  Implemented voice recognition (STT) and speech synthesis (TTS) modules to deliver natural language interactions, enhancing overall usability and engagement.

Shadow Realm – 2D Systems & AI Design
Project manager – Developer/Designer

Jan 2025 – Mar 2025

- Spearheaded the design and development of a 2D pixel art video game using Unity and C#, focusing on engaging gameplay mechanics and smooth performance.
- Created and animated detailed pixel art sprites, implementing core game logic for character movement, combat, and interactive elements.
- Applied agile development practices to iteratively test, refine, and optimize game features, ensuring a captivating player experience.

Roomie AI – Smart Living Ecosystem
Co-founder – AI Developer

Mar 2025 – Jul 2025

- Co-led the development of an AI-driven roommate and campus life management platform aimed at simplifying shared living through smart automation.
- Designed intelligent systems for expense tracking, grocery planning, and chore scheduling, ensuring fair distribution and task accountability.
- Built robust data pipelines to analyze user input and generate adaptive routines based on household habits and preferences.
- Developed and deployed a cross-platform mobile app using Flutter and Firebase, enabling real-time collaboration and offline functionality.
- Integrated intelligent reminders, push notifications, and scheduling logic to streamline communication and enhance daily coordination.

Maisonneuve College — **Montreal**
Technical Diploma in Computer Science — *Sept 2022 - May 2025*
Concentration in Application Development

Concordia University – Gina Cody Engineering School — **Montreal**
Degree in Computer Science — August 2025 - current

SKILLS

Software Engineering | Scalable Backend Architecture | Full-Stack Web Development | UI/UX & Mobile Design
AI & Machine Learning | Neural Network Implementation | RAG Systems | DevOps & Automated Deployment